

HoneyShield: A Multi-Layer Honeypot-Driven Threat Intelligence Framework for Detecting and Neutralizing Fake YONO APK Campaigns

Abstract

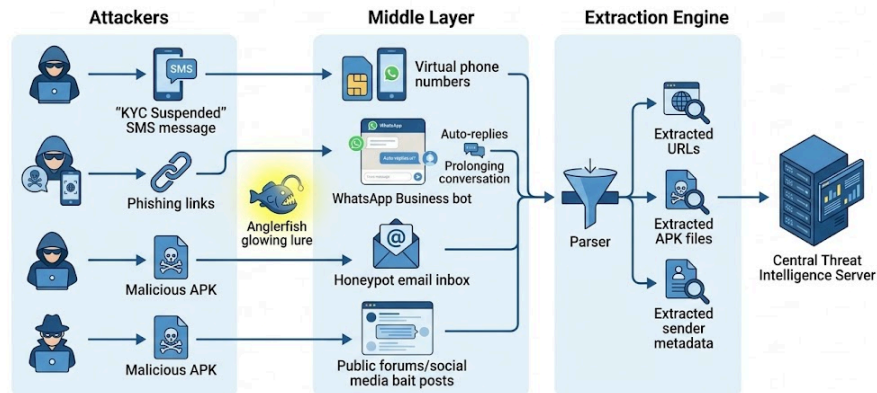
In the deep ocean, the anglerfish deploys a double deception: a bioluminescent lure draws prey close, and a concealed jaw snaps. HoneyShield mirrors this instinct — a two-stage honeypot architecture that lures fraudsters, then closes the trap when they act on what they believe they have stolen, combating the rising threat of fake YONO APKs targeting SBI customers.

The Lure (Multi-Channel Honeypot Collection): Virtual phone numbers are seeded across public forums with SBI-related bait content, attracting scammers who send fraudulent “KYC suspended” messages carrying malicious links and APKs. A WhatsApp Business API bot auto-responds to deepen engagement. Honeypot email addresses capture phishing emails; an automated parser extracts URLs, attachments, and sender metadata. All artifacts are routed to a centralized threat intelligence server.

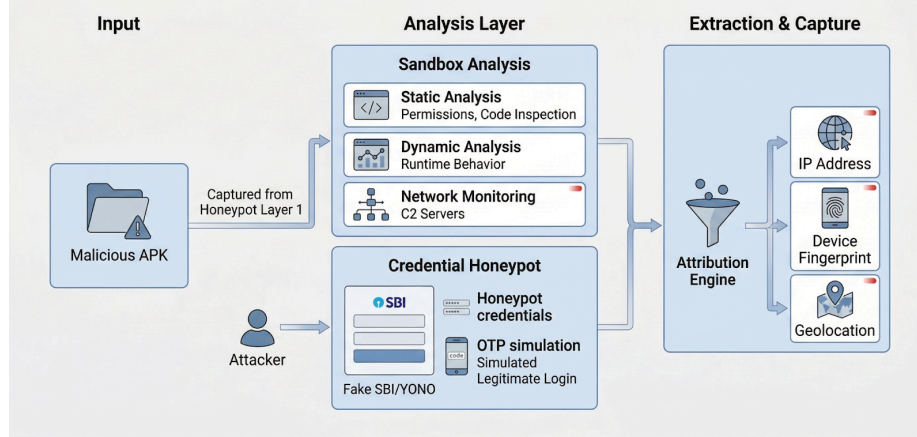
The Hidden Jaw (Sandbox Analysis & Credential Honeypot): Captured APKs are detonated in a fully isolated sandbox. Static and dynamic analysis exposes C2 servers, dangerous permissions, and live network connections the malware attempts. The app is then fed SBI-provisioned honeypot credentials linked to virtual numbers, with OTPs injected to simulate real SIM behaviour. When the attacker authenticates on the genuine YONO portal using these tagged credentials, the bank instantly captures their IP, device fingerprint, and geolocation for coordinated law enforcement takedown.

Ecosystem Response: Since fraudsters blast campaigns to thousands of real customers simultaneously, speed is critical. The moment a malicious URL hits our server, it is pushed to ISP-level DNS blocklists and Google Safe Browsing within minutes. Spoofed sender IDs are reported to TRAI via the Chakshu portal; APK hashes are shared with YONO’s backend to trigger on-device auto-alerts when unofficial SBI-branded apps are detected. A Fraud Dashboard aggregates all findings for continuous monitoring.

Honeypot Layer 1: Multi-Channel Lure & Scam Collection



Honeypot Layer 2: Sandbox Analysis & Credential Trap (Hidden Jaw)



HoneyShield Threat Intelligence Reporting & Ecosystem Response

